

Yijing Zhang

RESEARCH INTERESTS

AI for Scientific computing, Data-centric Machine Learning, Foundation Models.

CONTACT INFORMATION

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Website: <https://yijingz02.github.io/>

Github: <https://github.com/yijingz02/>

EDUCATION

Ph.D. in Computer Science. University of Wisconsin - Madison

Sep 2025 - Present

- Faculty Advisor: Prof. Misha Khodak.
- Selected courses taken: Machine Learning, Computational Mathematics, Optimization.

B.S. in Computer Science. University of Wisconsin - Madison

Sep 2021 - May 2024

- with Honor in the Major, Computer Science

PUBLICATIONS

Yijing Zhang, Nicholas Roberts, Tanya Marwah, Mikhail Khodak. *Breakeven complexity: A new perspective on neural partial differential equation solvers*. Preprint. In submission to Neurips 2026. [\[Paper\]](#) [\[Code\]](#) [\[Dataset\]](#)

Yijing Zhang, Dyah Adila, Changho Shin, Frederic Sala. *Personalize Your LLM: Fake it then Align it*. Annual Conference of the North American Chapter of the Association for Computational Linguistics(NAAACL) 2025 Findings. [\[Paper\]](#)

Yijing Zhang, Frederic Sala. *Methods for Domain-specific Fine-tuning for Generative Models*. Senior honor thesis for Honors in the Major, L&S Honors Program at University of Wisconsin - Madison, 2024. [\[Paper\]](#)

Lin Zhang, Shentong Mo, **Yijing Zhang**, Pedro Morgado. *Audio-Guided Visual Animation*. Oral presentation(Top 2.3%) at European Conference on Computer Vision(ECCV) 2024. [\[Paper\]](#) [\[Code\]](#)

Dyah Adila, Changho Shin, **Yijing Zhang**, Frederic Sala. *Can Language Models Safeguard Themselves, Instantly and For Free?* International Conference on Machine Learning(ICML) 2024 Workshop on NextGenAISafety. [\[Paper\]](#)

Dyah Adila, Changho Shin, **Yijing Zhang**, Frederic Sala. *Is Free Self-Alignment Possible?* Conference on Neural Information Processing Systems(NeurIPS) 2024 Workshop Foundation Model Interventions. [\[Paper\]](#)

Lin Zhang, Zefan Cai, Yufan Zhou, Shentong Mo, Jinhong Lin, Cheng-En Wu, Yibing Wei, **Yijing Zhang**, Ruiyi Zhang, Wen Xiao, Tong Sun, Junjie Hu, Pedro Morgado. *Scaling Up Audio-Synchronized Visual Animation: An Efficient Training Paradigm*. Preprint. [\[Paper\]](#)

AWARDS & HONORS

Summer Graduate Fellowship

April, 2026

Honors in the Major, Computer Science.

May, 2024

- Thesis: "Methods for Domain-specific Fine-tuning for Generative Model".
- University of Wisconsin - Madison, College of Letters & Science, L&S Honors Program

ACM ICPC 2021 NA Regional Contest: Team Rank 14.

2021

Driftless-star

2025 - Present

@University of Wisconsin - Madison

- Related Topic: AI for Scientific Computing, plasma physics, fusion
- Faculty advisors: Prof. Rogerio Jorge(PI), Prof. Misha Khodak(PI), Prof. Adelle Wright, Prof. Stephen Wright, Prof. Qin Li, Prof. Kyle Cranmer
- An end-to-end pipeline for transport-consistent stellarator optimization.
- Funded by Wisconsin Alumni Research Foundation and DOE Office of Fusion Energy Sciences (Award Numbers DE-SC0026040, DE-AC02-09CH11466 and DE-AC05-00OR22725).

KineticBench

July 2026 - Present

@University of Wisconsin - Madison

- Related Topic: AI for scientific computing, Kinetic theory
- Supervisor: Prof. Misha Khodak, Prof. Qin Li
- Building benchmark for kinetic theory and designing machine learning architecture for kinetic data.

Breakeven Complexity

Jun 2025 - Present

@University of Wisconsin - Madison

- Related Topic: AI for scientific computing, PDEs(partial differential equations)
- Supervisor: Prof. Misha Khodak
- Proposed an evaluation framework centered on breakeven complexity, a metric that counts the forward solves before a learned solver is cost-effective relative to an error-equivalent traditional solver.

CHAMELEON

May 2024 - Dec 2024

@University of Wisconsin - Madison

- Related Topic: NLP, Alignment, Personalization.
- Supervisor: Prof. Fred Sala.
- Utilizing AlignEZ approach for large-scale, low-resource personalization for LLMs.

Audio-guided Animation

Mar 2023 - May 2024

@University of Wisconsin - Madison

- Related Topic: Computer vision, Generative models, Audio-to-Video.
- Supervisor: Prof. Pedro Morgado.
- Focused on generating audio-video highly synchronized animation with guidance on audio.

AlignEZ

May 2024 - Sep 2024

@University of Wisconsin - Madison

- Related Topic: NLP, Alignment.
- Supervisor: Prof. Fred Sala.
- Focused on aligning pretrained language models without additional training.

Self-Flip-guided Classification

Sep 2024 - May 2025

@Brown University & University of Wisconsin - Madison

- Related Topic: NLP, Generative models.
- Supervisor: Prof. Stephen Bach & Prof. Fred Sala.
- Focus on a weak supervision approach that utilize self-flip to annotate weakly annotated text for classification.

Methods for Domain-specific Fine-tuning for Generative Model

Feb 2022 - May 2024

@University of Wisconsin - Madison

- Related Topics: NLP, Foundation models, Generative models, Fine-tuning, Data Efficiency
- Supervisor: Prof. Fred Sala.
- Served as independent research study for senior honor thesis.

- Focused on investigating the retrainability of synthetic datasets generated by fine-tuned generative models for domain-specific downstream classification tasks and the fine-tuning efficiency for generating higher-quality synthetic datasets.

EXPERIENCE	Research Assistant <i>@University of Wisconsin-Madison</i> • Work as research assistant under Prof. Fred Sala. • Collaborate with Prof. Stephen Bach from Brown University, we focus on self-flip-guided classification.	Sep 2024 - May 2025
SERVICE	Conference Reviewing Reviewer for ICML 2026, NeurIPS 2026.	
TEACHING	Teaching Assistant <i>CS220, University of Wisconsin-Madison</i>	Sep 2025 - Dec 2025
	Peer Mentor <i>CS400, University of Wisconsin-Madison</i>	Jan 2022 - May 2024